

PRANAV RAJESH CHARAKONDALA

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PROFESSIONAL SUMMARY

Forward deployed engineer and applied AI practitioner, MS Information Management (UIUC, May 2026), with 2+ years of industry experience building and deploying full-stack AI applications end to end, working directly with healthcare and enterprise clients to design, develop, and ship production systems, and mentoring junior engineers across healthcare, financial services, and research environments.

PROFESSIONAL EXPERIENCE

CENTER FOR HEALTH INFORMATICS, UIUC | Champaign, IL

Lead AI Engineer | Agentic AI & LLM Platform Engineering

May 2025 – May 2026

- Embedded with **WHO stakeholders as the client**, independently scoping and architecting a zero-to-one production platform for health misinformation detection, leading 12 student researchers and delivering technical enablement to WHO representatives
- Fine-tuned **Mistral 7B using QLoRA** (PEFT, rank-16 LoRA, 4-bit quantization) on 500 curated health domain transcripts, improving classification accuracy from 71% to 84% and DEBUNKING F1 from 52% to 78%
- Established an **eval and observability framework** benchmarking multiple LLM providers across precision, recall, F1, and latency with YAML configs and confidence histograms, running feedback loops across 40+ experimental runs
- Shipped an **analyst-facing AI platform** with a Next.js 14 frontend enabling non-technical public health researchers to independently run AI workflows, translating complex model outputs into accessible insights

DELOITTE USI | Bangalore, India

Software Engineer | Financial Services Platform

Jun 2022 – Jul 2024

- Led **embedded client delivery** across UAT cycles in fast-paced, ambiguous environments, joining client calls to hear requirements directly, translating business needs into production-grade solutions, and resolving 60+ integration defects under tight timelines
- Delivered **50+ production features and custom integrations** in Java and JavaScript/Node.js for a U.S. wealth management platform supporting 12,000+ financial advisors, migrating from multiple legacy systems to a unified Salesforce FSC platform
- Reviewed **technical design documents and code** across multiple workstreams, mentoring 2 junior analysts on coding standards and design techniques, and producing technical documentation to standardize deployment workflows
- Automated resolution of **200+ Salesforce-DB2 data discrepancies** using SQL and Snowflake-based anomaly detection, originating and owning the solution end to end with no predefined approach, cutting investigation time by 40%

DRONGO AI (Early-Stage AI Startup) | Bangalore, India

Machine Learning Intern | Medical AI & Real-Time ML Systems

Jan 2022 – Jun 2022

- Trained **medical AI models** using UNet and PraNet on dental radiographs for osteoporosis detection alongside dental specialists, achieving a 10% Dice coefficient improvement through CUDA-optimized PyTorch training pipelines
- Developed a **real-time inference pipeline** using YOLOv4 and PP-OCR achieving 95% OCR accuracy at 30 FPS with CUDA-based optimization, containerized with Docker and deployed on AWS EC2

PROJECT EXPERIENCE

ReconAI – Production Agentic AI System for Enterprise Integration Diagnosis

Mar 2026 – Apr 2026

- Architected a **stateful 4-agent LangGraph system** with autonomous branching, retries, self-correction, and proposer-critic-judge reasoning, implementing tool orchestration across 6 external systems and dual RAG patterns, achieving 70% RCA at \$0.006/incident
- Owned the **end-to-end production platform** from first commit to deployment: FastAPI, React 18, TypeScript, Supabase PostgreSQL, SSE streaming, replacing 3-hour manual workflows with automated AI workflows for RCA in under 15 seconds, live at recon-ai-iota.vercel.app

Swagath Central – Production Operations Platform for Multi-Location Business (Freelance)

Feb 2026 – May 2026

- Led a **0-1 embedded client engagement** with no documentation, decoding a paper-based operational system from receipts, then shipped a production pipeline with reconciliation, anomaly detection, and automated AI workflows using Llama 3.3 70B
- Recovered **₹50,000 in 3 months** by diagnosing multiple production bugs affecting live financial data, including a database concurrency deadlock and a silent data-integrity failure, turning each fix into reusable integration patterns and documentation for the incoming team

TECHNICAL SKILLS

Programming & Languages: Python (Advanced); TypeScript; JavaScript/Node.js; Java; SQL; OOP; RESTful APIs; Distributed Systems

AI & Agentic Systems: LLMs & Gen AI; Agentic AI; Multi-Agent Systems; LangGraph; LangChain; RAG; FAISS; pgvector; Claude & OpenAI APIs; Prompt Engineering; Fine-Tuning; QLoRA; Eval Pipelines; Observability; Human-in-the-Loop

Backend & Infrastructure: FastAPI; Docker; Kubernetes; AWS (EC2, S3); PostgreSQL; Redis; Celery; CI/CD (GitHub Actions); Git; Supabase

Frontend & Full-Stack: React; Next.js; TypeScript; Tailwind CSS; SSE Streaming; REST API Design

EDUCATION

University of Illinois Urbana – Champaign, Champaign, Illinois

Aug 2024 - May 2026

Master of Science Information Management, **GPA 3.86/4.00**

PES University, Bangalore, India

Aug 2018 - May 2022

Bachelor of Technology Electronics and Communications Engineering, **GPA 7.72/10.00**